WEEK – 2

2. Load AirPassengers dataset in R and visualize the data using line chart that shows increase in air passengers over given time period.

AIM:

To visualize the increase in air passengers over a given time period using line chart.

DESCRIPTION:

The built-in AirPassengers dataset is loaded and analyzed. This dataset contains the monthly total number of international airline passengers (in thousands) from 1949 to 1960.

First, the summary() function is used to display basic statistics of the dataset, including minimum, maximum, median, mean, and quartiles, which helps understand the range and distribution of passenger numbers. The paste() function concatenates text strings in R. The class() function confirms that AirPassengers is a time-series object, meaning it is ordered by time and can be used for time-based analysis.

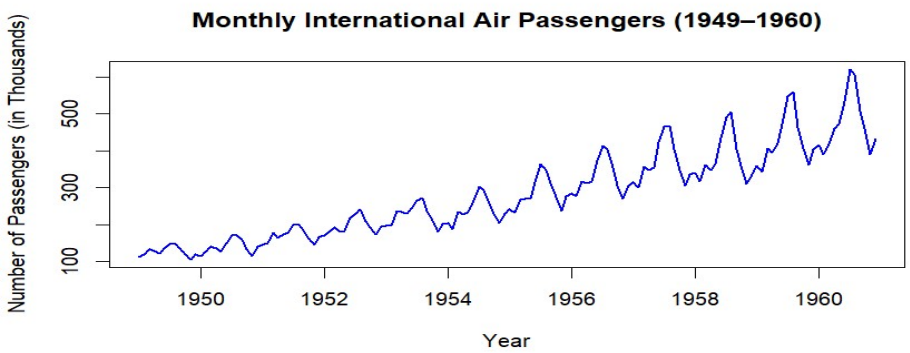
A line chart is plotted to visualize the trend in monthly passenger numbers over the years. This visualization helps in understanding **trends, patterns, and seasonal variations** in the airline passenger data over the 12-year period.

PROGRAM:

```
# Step 1: Load the dataset
data("AirPassengers")

# Step 2: Display basic details of the dataset
print("Summary of AirPassengers dataset:")
summary(AirPassengers)

# Step 3: Check the class (time-series object)
print(paste("Class of dataset:", class(AirPassengers)))

# Step 4: Visualize using a line chart
plot(AirPassengers,
     main = "Monthly International Air Passengers (1949–1960)",
     xlab = "Year",
     ylab = "Number of Passengers (in Thousands)",
     col = "blue",
     type = "l",   # Line chart
     lwd = 2)      # Line width
```

OUTPUT:

```
 Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 104.0   180.0   265.5   280.3   360.5   622.0

"Class of dataset: ts"
```



WEEK – 4

4. a) Load iris dataset in R, visualize the data using different Box plots including group by option and also use color palette to represent species.

AIM:

To visualize the iris data using different Box plots including group and also use color palette to represent species.

DESCRIPTION:

The iris dataset is loaded into the R environment. A summary of the dataset is displayed, showing statistical information such as minimum, maximum, median, and quartiles for each numeric variable. A boxplot of **Sepal Length** grouped by species is created. Different colors (lightblue, salmon, lightgreen) are used to distinguish the three species, and a dark blue border is applied to the boxes. A second boxplot is generated for **Petal Width** grouped by species, again using a different set of custom colors (skyblue, orange, lightpink) to highlight the differences among species. A color palette is generated using the rainbow() function to create three distinct colors. A boxplot of **Sepal Width** grouped by species is drawn using the rainbow color palette. This helps visualize how Sepal Width varies among the three iris species.

Overall, the program uses multiple boxplots and different color schemes to compare the flower measurements across species.

PROGRAM:

```
# Step 1: Load iris dataset
data("iris")

# Step 2: View dataset summary
summary(iris)

# Step 3: Boxplot of Sepal Length grouped by Species with color
boxplot(Sepal.Length ~ Species,
        data = iris,
        main = "Sepal Length by Species",
        xlab = "Species",
        ylab = "Sepal Length (cm)",
        col = c("lightblue", "salmon", "lightgreen"),
        border = "darkblue")

# Step 4: Boxplot of Petal Width grouped by Species with different colors
boxplot(Petal.Width ~ Species,
```

```
      data = iris,
      main = "Petal Width by Species",
      xlab = "Species",
      ylab = "Petal Width (cm)",
      col = c("skyblue", "orange", "lightpink"),
      border = "black")
```

 # Step 5: Use RColorBrewer for color palette

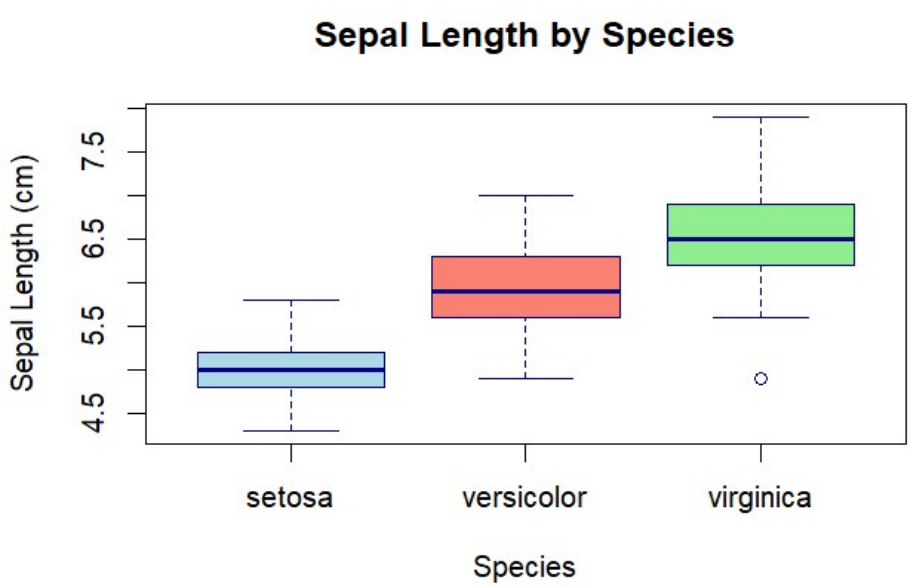
```
palette <- rainbow(3)
```

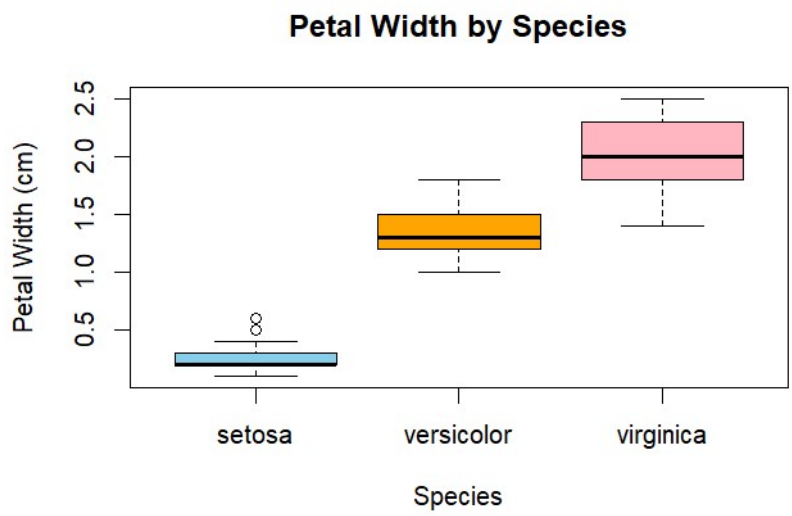
 # Step 6: Boxplot for Sepal Width grouped by Species using palette

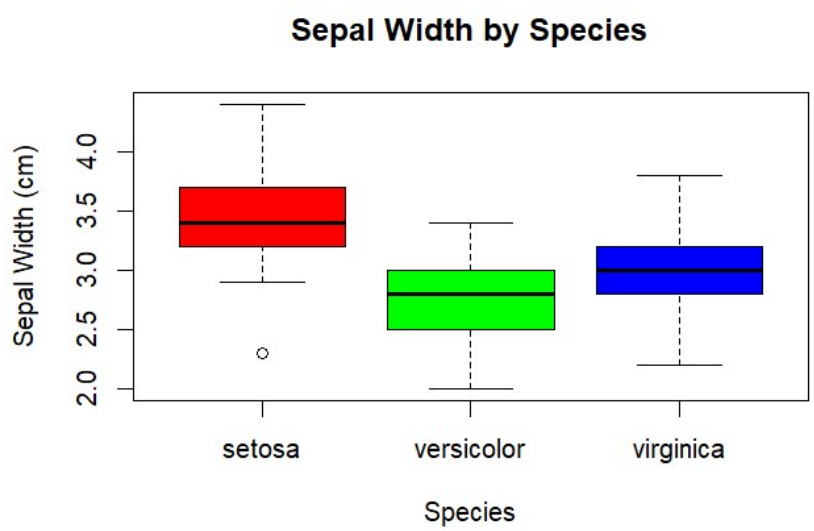
```
boxplot(Sepal.Width ~ Species,
      data = iris,
      main = "Sepal Width by Species",
      col = palette,
      xlab = "Species",
      ylab = "Sepal Width (cm)")
```

OUTPUT:

```
 Sepal.Length    Sepal.Width     Petal.Length    Petal.Width
Min.   :4.300   Min.   :2.000   Min.   :1.000   Min.   :0.100
1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300
Median :5.800   Median :3.000   Median :4.350   Median :1.300
Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199
3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800
Max.   :7.900   Max.   :4.400   Max.   :6.900   Max.   :2.500
       Species
setosa    :50
versicolor:50
virginica :50
```




Petal Width by Species
Petal Width (cm)
0.5
1.0
1.5
2.0
2.5
setosa
versicolor
virginica
Species


Sepal Width by Species
Sepal Width (cm)
2.0
2.5
3.0
3.5
4.0
setosa
versicolor
virginica
Species

b) Load air quality dataset in R and visualize air quality parameters using box plots.

AIM:

To visualize air quality parameters on air quality dataset using box plots.

DESCRIPTION:

The program analyzes and visualizes the airquality dataset using boxplots in R. The variables include Ozone concentration, Solar Radiation, Wind speed, and Temperature. The built-in airquality dataset is loaded into R. A summary of the dataset is displayed, showing statistical details such as minimum, median, mean, and maximum values for each variable. This provides an initial understanding of the data distribution.

Four separate boxplots are created for Ozone, Solar Radiation, Wind speed, and Temperature. **par(mfrow = c(2,2))** is used to divide the plotting area into a 2 × 2 grid, allowing four boxplots to be displayed together in one window for easy comparison. Each variable is plotted with a different color to visually distinguish them. Ozone → light blue, Solar Radiation → light green, Wind → light yellow, Temperature → light pink. The boxplots show the spread of data, median values, and presence of any outliers for each parameter.

Overall, the program visually summarizes the variability in air quality parameters, helps identify outliers, and allows comparison of the distribution of Ozone, Solar Radiation, Wind, and Temperature across the dataset.

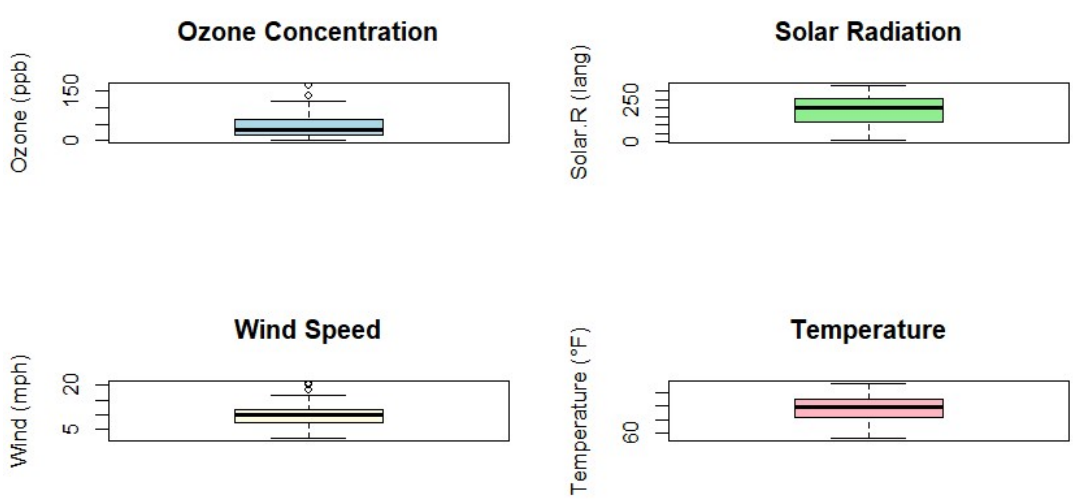
PROGRAM:

```
# Step 1: Load airquality dataset
data("airquality")

# Step 2: View dataset summary
summary(airquality)

# Step 3: Boxplots for Ozone, Solar.R, Wind, and Temp
par(mfrow = c(2, 2))  # 2x2 layout

boxplot(airquality$Ozone,
        main = "Ozone Concentration",
        ylab = "Ozone (ppb)",
        col = "lightblue")

boxplot(airquality$Solar.R,
        main = "Solar Radiation",
        ylab = "Solar.R (lang)",
        col = "lightgreen")
```

```
boxplot(airquality$Wind,
        main = "Wind Speed",
        ylab = "Wind (mph)",
        col = "lightyellow")

boxplot(airquality$Temp,
        main = "Temperature",
        ylab = "Temperature (°F)",
        col = "lightpink")

par(mfrow = c(1, 1))  # Reset layout
```

OUTPUT:

```
     Ozone           Solar.R           Wind             Temp
 Min.   :  1.00   Min.   :  7.0   Min.   : 1.700   Min.   :56.00
 1st Qu.: 18.00   1st Qu.:115.8   1st Qu.: 7.400   1st Qu.:72.00
 Median : 31.50   Median :205.0   Median : 9.700   Median :79.00
 Mean   : 42.13   Mean   :185.9   Mean   : 9.958   Mean   :77.88
 3rd Qu.: 63.25   3rd Qu.:258.8   3rd Qu.:11.500   3rd Qu.:85.00
 Max.   :168.00   Max.   :334.0   Max.   :20.700   Max.   :97.00
 NA's   :37       NA's   :7
     Month            Day
 Min.   :5.000   Min.   : 1.0
 1st Qu.:6.000   1st Qu.: 8.0
 Median :7.000   Median :16.0
 Mean   :6.993   Mean   :15.8
 3rd Qu.:8.000   3rd Qu.:23.0
 Max.   :9.000   Max.   :31.0
```



WEEK – 5

5. Visualize iris dataset using simple scatter, multivariate scatter plot and also visualize scatter plot matrix to visualize multiple variables across each other.

AIM:

To visualize iris dataset using simple scatter, multivariate scatter plot and visualize multiple variables across each other.

DESCRIPTION:

The program loads the built-in iris dataset and visualizes it using different scatter plot techniques. Initially, the dataset is displayed to observe its structure and values. A simple scatter plot is created between Sepal Length and Sepal Width, where each data point represents a flower and colors are used to distinguish different species. A legend is added to improve clarity and interpretation of the species-wise distribution.

Next, a multivariate scatter plot is drawn between Sepal Length and Petal Length, again using color to represent species, which helps in comparing relationships across multiple variables simultaneously. Finally, a scatter plot matrix is generated for all four numerical attributes of the iris dataset, allowing observation of pairwise relationships and patterns among variables. Overall, the program helps in identifying trends, correlations, and species separation through effective graphical visualization.

PROGRAM:

```
# Load iris dataset
data("iris")
print(iris)

# Simple scatter plot: Sepal Length vs Sepal Width
plot(iris$Sepal.Length, iris$Sepal.Width,
    main = "Scatter Plot: Sepal Length vs Sepal Width",
    xlab = "Sepal Length (cm)",
    ylab = "Sepal Width (cm)",
    col = iris$Species,
    pch = 19)

# Add legend
legend("topright", legend = levels(iris$Species),
       col = 1:3, pch = 19)

# Multivariate scatter plot: Sepal Length vs Petal Length, colored by Species
plot(iris$Sepal.Length, iris$Petal.Length,
```

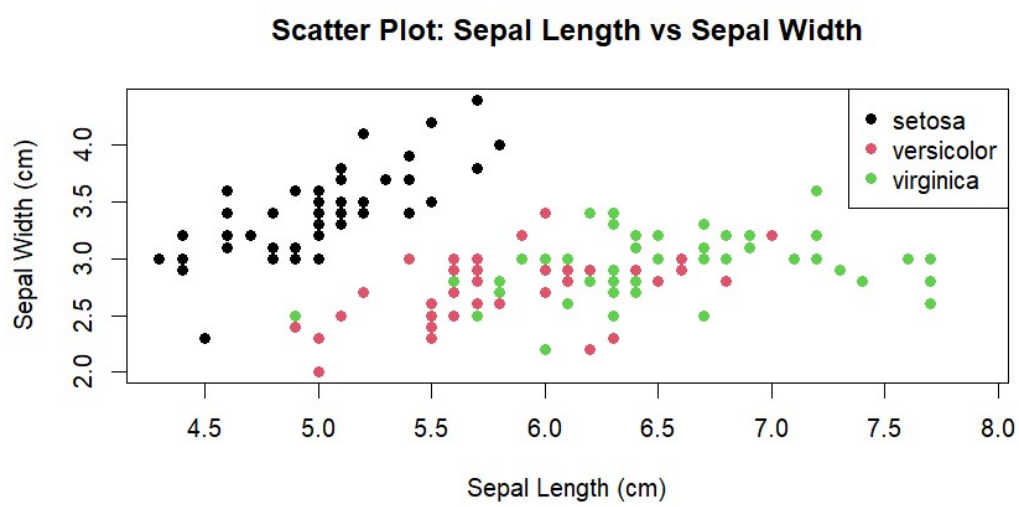
```
    main = "Multivariate Scatter: Sepal vs Petal Length",
    xlab = "Sepal Length (cm)",
    ylab = "Petal Length (cm)",
    col = iris$Species,
    pch = 19)
```

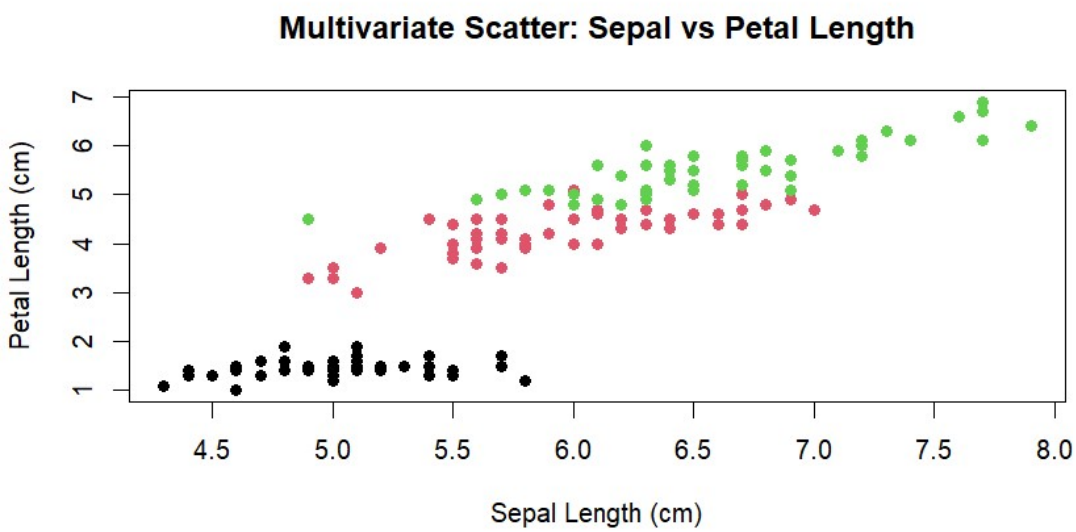
 # Scatter plot matrix

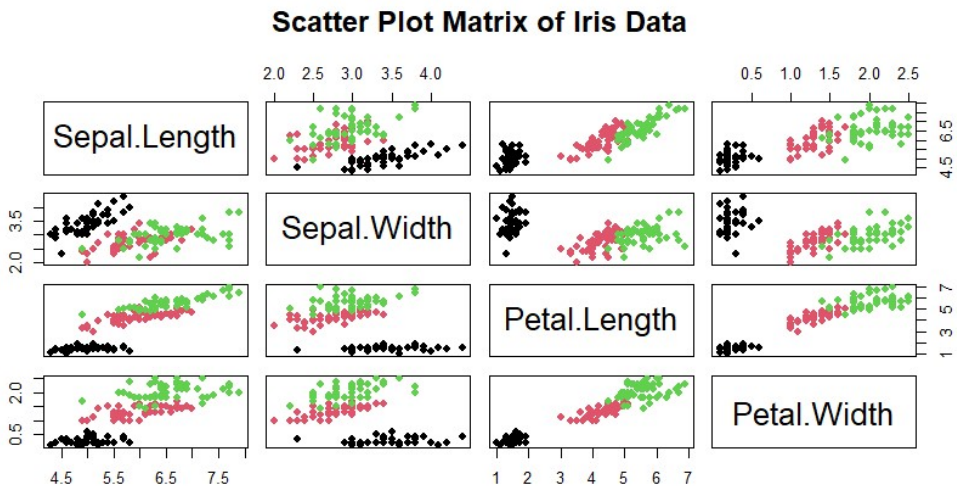
```
pairs(iris[1:4],
     main = "Scatter Plot Matrix of Iris Data",
     col = iris$Species,
     pch = 19)
```

OUTPUT:

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
```




Multivariate Scatter: Sepal vs Petal Length
Petal Length (cm)
Sepal Length (cm)
1
2
3
4
5
6
7
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0


Scatter Plot Matrix of Iris Data
Sepal.Length
Sepal.Width
Petal.Length
Petal.Width
2.0 2.5 3.0 3.5 4.0
0.5 1.0 1.5 2.0 2.5
4.5 6.5
2.0 3.5
1 3 5 7
0.5 2.0
4.5 5.5 6.5 7.5
1 2 3 4 5 6 7

WEEK – 7

7. Load Hair Eye Color dataset in R and plot categorical data using mosaic plot.

AIM:

To Load Hair Eye Color dataset in R and plot categorical data using mosaic plot.

DESCRIPTION:

The program uses the built-in HairEyeColor dataset in R, which represents the frequency distribution of hair color, eye color, and gender. The dataset is first viewed in its original table format. It is then optionally converted into a data frame (hair_eye_df) for easier manipulation or analysis.

A **mosaic plot** is created to visualize the relationship between **hair color** and **eye color**. The mosaic plot displays the proportion of individuals for each combination of hair and eye color, where the size of each rectangle is proportional to the frequency of that combination. Colors are used to enhance the visualization, making patterns and associations between hair and eye colors easier to interpret.

PROGRAM:

```
# Load the built-in HairEyeColor dataset
data("HairEyeColor")

# View the dataset
HairEyeColor

# Convert the table to a data frame (optional)
hair_eye_df <- as.data.frame(HairEyeColor)

# Create a mosaic plot
mosaicplot(HairEyeColor,
        main = "Mosaic Plot of Hair Color vs Eye Color",
        xlab = "Hair Color",
        ylab = "Eye Color",
        color = TRUE)
```

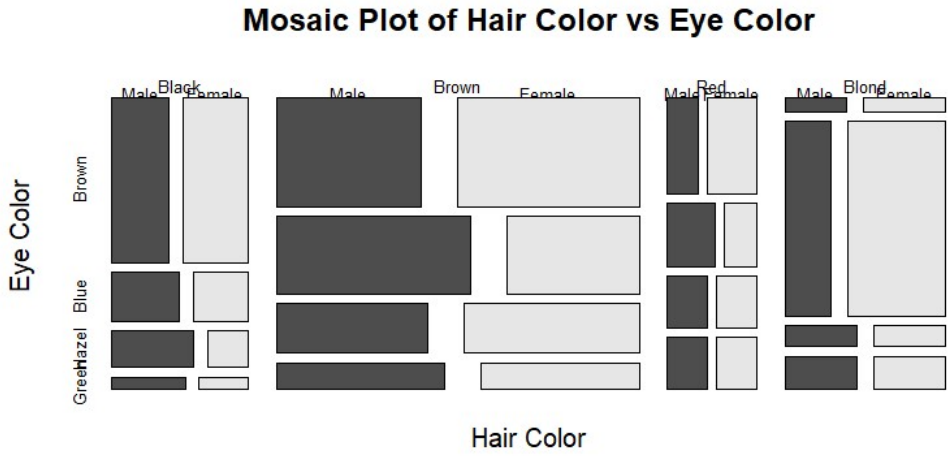
OUTPUT:

Sex = Male

Hair	Eye Brown	Blue	Hazel	Green
Black	32	11	10	3
Brown	53	50	25	15
Red	10	10	7	7
Blond	3	30	5	8

, , Sex = Female

Hair	Eye Brown	Blue	Hazel	Green
Black	36	9	5	2
Brown	66	34	29	14
Red	16	7	7	7
Blond	4	64	5	8



WEEK – 8

8. Load mtcars dataset in R and visualize data using heat map.

AIM:

To Load mtcars dataset in R and visualize data using heat map.

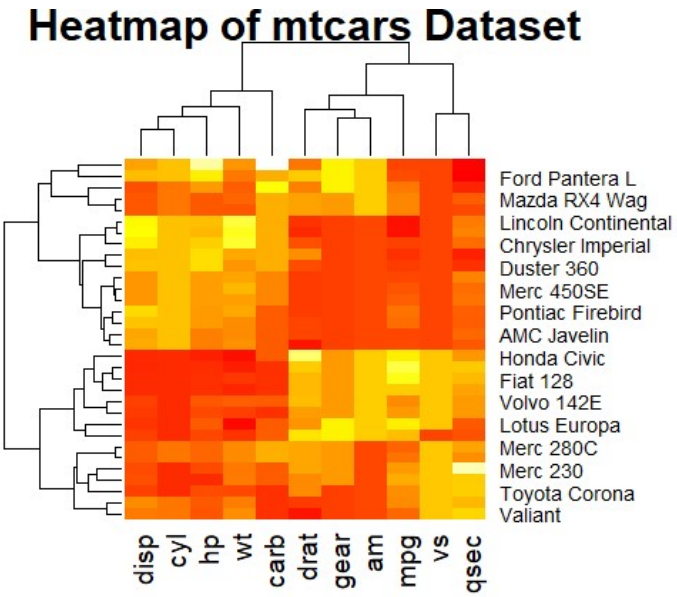
DESCRIPTION:

The program loads the built-in **mtcars** dataset and displays its summary to understand the basic statistics of each variable. The data is then **normalized using the scale() function** so that all variables are brought to a common scale, which improves heatmap interpretation. A **heatmap** is generated using base R to visualize patterns, similarities, and variations among cars and variables. The color intensity represents the magnitude of values, making it easier to identify clusters and relationships within the dataset.

PROGRAM:

```
# Load the dataset
data("mtcars")

#Summary of dataset
summary(mtcars)

# Normalize the data (optional for better heatmap visualization)
mtcars_scaled <- scale(mtcars)

# Heatmap using base R
heatmap(mtcars_scaled,
        main = "Heatmap of mtcars Dataset",
        col = heat.colors(256),
        scale = "none")
```

OUTPUT:

```
     mpg             cyl             disp           hp              drat
 Min.   :10.40   Min.   :4.000   Min.   : 71.1   Min.   : 52.0   Min.   :2.760
 1st Qu.:15.43   1st Qu.:4.000   1st Qu.:120.8   1st Qu.: 96.5   1st Qu.:3.080
 Median :19.20   Median :6.000   Median :196.3   Median :123.0   Median :3.695
 Mean   :20.09   Mean   :6.188   Mean   :230.7   Mean   :146.7   Mean   :3.597
 3rd Qu.:22.80   3rd Qu.:8.000   3rd Qu.:326.0   3rd Qu.:180.0   3rd Qu.:3.920
 Max.   :33.90   Max.   :8.000   Max.   :472.0   Max.   :335.0   Max.   :4.930
```

```
      wt              qsec            vs                 am                gear
 Min.   :1.513   Min.   :14.50   Min.   :0.0000   Min.   :0.0000   Min.   :3.000
 1st Qu.:2.581   1st Qu.:16.89   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:3.000
 Median :3.325   Median :17.71   Median :0.0000   Median :0.0000   Median :4.000
 Mean   :3.217   Mean   :17.85   Mean   :0.4375   Mean   :0.4062   Mean   :3.688
 3rd Qu.:3.610   3rd Qu.:18.90   3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:4.000
 Max.   :5.424   Max.   :22.90   Max.   :1.0000   Max.   :1.0000   Max.   :5.000
     carb
 Min.   :1.000
 1st Qu.:2.000
 Median :2.000
 Mean   :2.812
 3rd Qu.:4.000
 Max.   :8.000
```